

APPLIED GLACIOLOGY

Assignment 1: Glaciology and Hydropower



Figure 1: Silvrettagletscher, Fluchthorn, Piz Buin v. W. aus 2800 m, 1925 [9]

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Introduction

This assignment focuses on the analysis and modelling of glacier mass balance within the framework of the course Applied Glaciology – Glaciology and Hydropower. The aim is to quantify past changes of the Silvretta Glacier, calibrate a simple temperature-index melt model, and apply it to reconstruct and project the glacier's evolution.

The work is divided into five interconnected tasks. In Task 1, the geodetic method is applied to determine the glacier's specific mass change rate between 1994 and 2022 using Digital Elevation Models (DEMs) and glacier masks. Task 2 involves selecting a representative meteorological station and retrieving long-term homogenized temperature and precipitation data from MeteoSwiss. These data serve as the basis for the Degree Day Factor (DDF) calibration in Task 3, where a simple melt model is tuned to reproduce the observed glacier mass change. Task 4 employs the calibrated model to reconstruct the annual glacier mass balance from 1935 to 2024, allowing the identification of long-term trends and climatic influences. Finally, Task 5 uses volume–area scaling and the reconstructed mass balance to project the future evolution of the glacier and estimate when it will lose half of its current volume.

Throughout the assignment, simplifying assumptions are made regarding glacier geometry, temperature lapse rates, and precipitation distribution. Despite these simplifications, the analysis provides valuable insight into glacier-climate interactions, the response of the Silvretta Glacier to climatic forcing, and the applicability of empirical melt models in glaciological studies.

1 Geodetic glacier mass balance

1.1 Question 1:

How large was the specific mass change rate experienced by the glacier during the period? Express your result in units of "m water equivalent per year" (m w.e. a⁻¹), and make sure to explain what assumptions you did for computing this number.

1.1.1 Fundamentals

To solve the task, the following input data and assumptions were provided or adopted for the analysis:

- **Digital Elevation Models (DEMs):** Surface elevation data of the Silvretta Glacier for the years 1994 and 2022. The DEMs provide elevation values on a regular grid and are used to compute elevation differences and, subsequently, the glacier volume change.
- **Glacier masks:** Binary raster files for the years 1994 and 2022 indicating whether a grid cell is covered by ice (1.0) or not (0.0). These masks ensure that only glacierized areas are considered in the calculations.
- **Time period:** The time span between the two DEMs is $\Delta t = 2022 - 1994 = 28$ years.
- **Density of glacier ice:** The density of glacier ice is assumed to be constant at $\rho_{\text{ice}} = 850 \text{ kg/m}^3$. This value represents an average density typically used for temperate glaciers.[4]
- **Density of water:** For the conversion of mass change to the unit water equivalent, the density of water is assumed to be $\rho_{\text{water}} = 1000 \text{ kg/m}^3$.
- **Additional assumptions:** It is assumed that the spatial resolution and coordinate reference systems of both DEMs and glacier masks are identical, ensuring a consistent cell-by-cell comparison. Furthermore, it is assumed that there is no systematic elevation offset between the two DEM datasets.

1.1.2 Methodology

The following calculation steps were executed to solve the task:

1. Difference in elevation of ice-covered cells:

To determine the change in elevation between 1994 and 2022, the glacier masks were used to isolate the ice-covered regions in both years. Each mask represents a binary grid of 201 rows and 324 columns. Each cell value of 1.0 corresponds to glacier-covered terrain and each value of 0.0 to ice-free areas. This ensures that only glacierized regions are considered when computing elevation differences.

In order to capture both glacier retreat and potential advance, a combined mask was generated. This mask includes all grid cells classified as glacier in either 1994 or 2022, thus accounting for all zones that experienced changes in glacier extent. The elevation change

for each grid cell was then derived from the Digital Elevation Models (DEMs) as following:

$$\Delta h = h_{2022} - h_{1994} \quad (1.1)$$

2. Calculation of volume change:

The net change in glacier volume ΔV between 1994 and 2022 was obtained by integrating the elevation differences over all glacier-covered grid cells. Only cells defined by the combined glacier mask were included, ensuring that both retreat and advance areas were considered. The elevation change Δh for each cell was multiplied by the corresponding cell area A_{cell} and the resulting products were summed for all glacier cells:

$$\Delta V = \sum_{\text{glacier cells}} (\Delta h \cdot A_{\text{cell}}) \quad (1.2)$$

This procedure yields the net change in glacier volume over the observed period.

3. Calculation of mass change:

The net change in mass of the glacier Δm was determined by converting the calculated volume change into mass using the assumed density of glacier ice. This relationship is expressed as follows:

$$\Delta m = \Delta V \cdot \rho_{\text{ice}} \quad (1.3)$$

where ρ_{ice} represents the density of glacier ice, assumed to be 850 kg/m^3 (see Section 1.1.1). The resulting value Δm quantifies the overall loss or gain of glacier mass over the studied period.

4. Calculation of glacier area:

The total glacier area A for the years 1994 and 2022 was determined by summing all grid cells identified as glacier-covered within the respective glacier masks and multiplying the number of glacier cells by the area of a single grid cell A_{cell} . This can be expressed as:

$$A_{\text{cell}} = 100 \text{ m}^2 \quad (1.4)$$

$$A = \sum_{\text{glacier cells}} A_{\text{cell}} \quad (1.5)$$

Subsequently, the average glacier area A_{avg} over the observation period was computed as the mean of the glacier areas from 1994 and 2022:

$$A_{\text{avg}} = \frac{A_{1994} + A_{2022}}{2} \quad (1.6)$$

This average area is later used to normalize the change of mass and derive the specific mass balance rate.

5. Calculation of specific mass change rate:

The specific mass change rate quantifies the amount of glacier mass that is lost or gained per year and per unit area. It is expressed in meters water equivalent per year (m w.e. a^{-1}). The value is obtained by dividing the total glacier mass change by the product of the water density, the average glacier area, and the time interval between the two DEMs:

$$\text{Specific mass change rate} = \frac{\Delta m}{\rho_{\text{water}} \cdot A_{\text{avg}} \cdot \Delta t} \quad (1.7)$$

where Δm is the total mass change of the glacier [kg], $\rho_{\text{water}} = 1000 \text{ kg/m}^3$ is the density of water, A_{avg} is the mean glacier area over the studied period [m^2], and $\Delta t = 28$ years represents the time span from 1994 to 2022.

1.1.3 Results

Between the years of 1994 and 2022, the studied glacier experienced a significant reduction in both volume and mass. Calculations indicate that the glacier's volume decreased by approximately $\Delta V = 83'071.8 \cdot 10^3 \text{ m}^3$, corresponding to a substantial mass loss of about $\Delta m = 70'611 \cdot 10^6 \text{ kg}$.

The extent of the glacier also decreased during this period. In 1994, the glacier area was $A_{1994} = 3'012 \cdot 10^3 \text{ m}^2$, while in 2022 it has shrunk to $A_{2022} = 2'326 \cdot 10^3 \text{ m}^2$, resulting in an average glacier area of approximately $A_{\text{avg}} = 2'669 \cdot 10^3 \text{ m}^2$ for the period considered.

Based on these changes, the specific mass change rate, which describes the annual mass loss per unit area, was calculated to be approximately $0.9448 \text{ m w.e. a}^{-1}$. These results clearly illustrate the considerable ice melt and retreat of the glacier over the past decades.

2 Meterological data

2.1 Question 2:

Retrieve a homogenized temperature and precipitation time series in monthly resolution for a meteorological station that you consider to be representative for the glacier. The time series shall cover the period since 1935, since that will be needed later. Note that only a few stations exist with such long-term time series. Which data did you choose and why?

Make sure to state the url from which you downloaded the data, and to discuss the choice of the station that you selected.

2.1.1 Fundamentals

The Silvretta Glacier is situated in the north east of the canton Graubünden. It extends from its highest point of 3079 m a.s.l. to its lowest point of 2467 m a.s.l.. Near the glacier are various measuring stations of the Swiss National Basic Climatological Network (NBCN). Temperature and precipitation records from NBCN stations have been fully digitized and homogenized from their start of measurement. Homogenization corrects artificial shifts caused by station relocations, instrument changes, or environmental alterations, ensuring historical climate data are consistent with today's measurement conditions. The following four stations have been inspected further as they cover the time period since 1935 and are in a reasonable distance to the Silvretta Glacier: [7]

- Davos (1594 m a.s.l.)
- Bad Ragaz (497 m a.s.l.)
- Samedan (1709 m a.s.l.)
- Segl-Maria (1804 m a.s.l.)

2.1.2 Methodology

To select the most suitable station, the norm value charts for temperature, precipitation, and sunshine duration from 1991 to 2020 were considered.[7]

In Figure 2.1, the mean annual temperature chart is shown. On the Silvretta Glacier, an average temperature between -3°C and 0°C is observed. In Bad Ragaz, the average temperature ranges from 10°C to 12°C . This higher temperature is mainly due to the large difference in elevation, with Bad Ragaz located at 497 m a.s.l. and the Silvretta Glacier at mean elevation of 2770 m a.s.l.. The other three locations (Davos, Samedan, and Segl-Maria) have average temperatures between 0°C and 2°C , which represent the glacier conditions more accurately than Bad Ragaz. These three villages are located within a height range of approximately 210 metres and sit around 1'000 metres below the glacier.

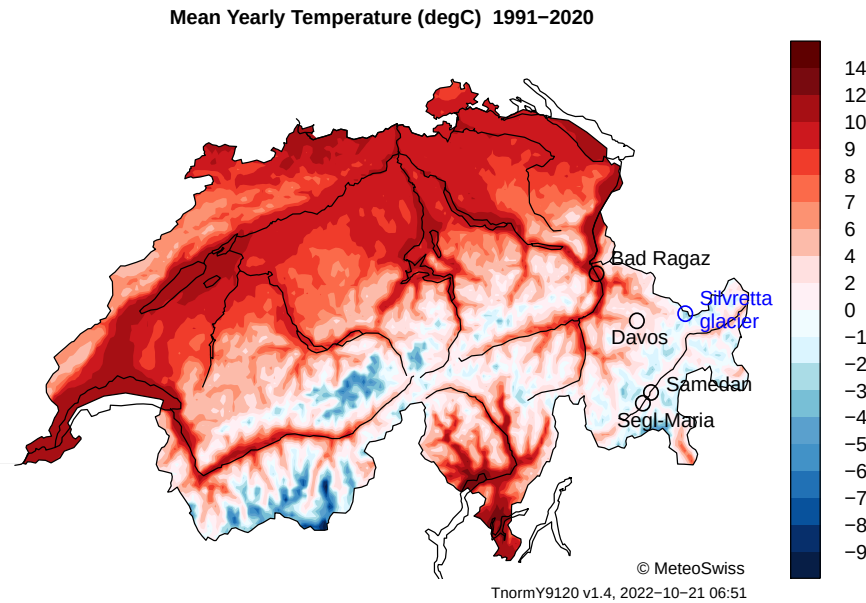


Figure 2.1: Mean yearly temperature chart, 1991 to 2020 [7]

When comparing the mean annual precipitation in Figure 2.2, Davos with values between 1100 mm and 1700 mm fits best the conditions of the Silvretta Glacier, where an average precipitation between 1500 mm and 1700 mm was observed. Segl-Maria and Bad Ragaz show lower precipitation values, ranging from 900 mm to 1300 mm, while Samedan records even less, between 700 mm and 1100 mm.

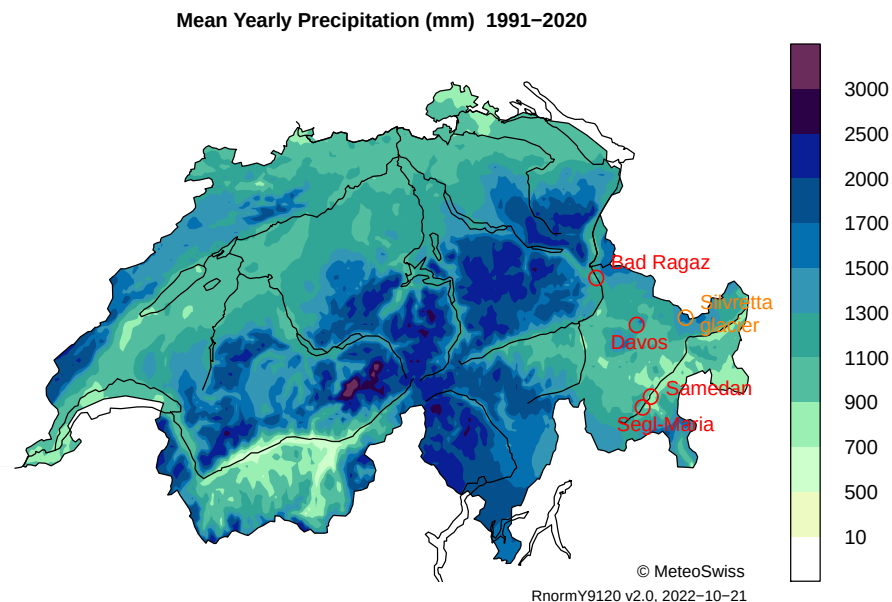


Figure 2.2: Mean yearly percipitation chart, 1991 to 2020 [7]

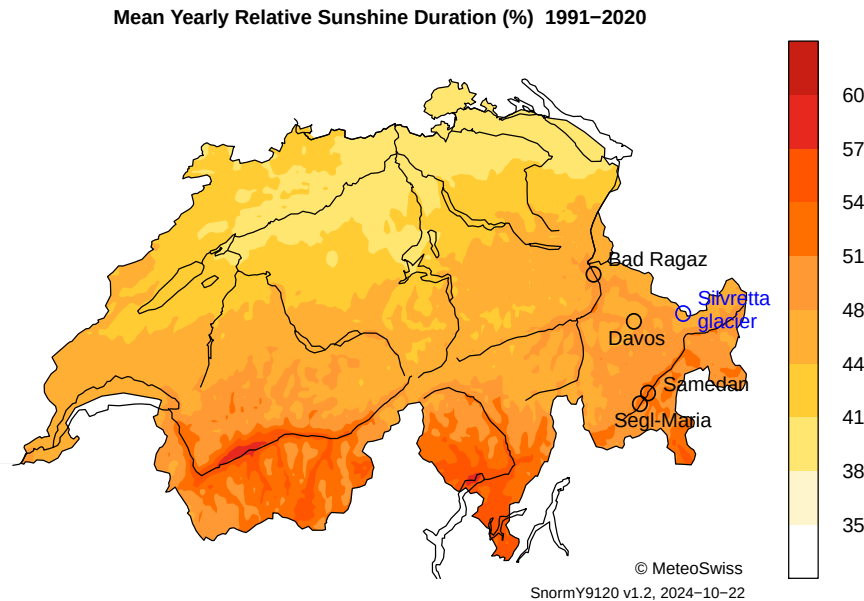


Figure 2.3: Mean yearly relative sunshine duration chart, 1991 to 2020 [7]

The mean yearly relative sunshine duration (Figure 2.3) further distinguishes these locations. The Silvretta Glacier records relative sunshine durations between 44% and 48%, similar to Davos, which lies in a valley surrounded by high mountains that limit direct sunlight. Bad Ragaz, by contrast, reaches slightly higher values of 48% to 51%, due to its lower altitude and more open valley geometry. The highest relative sunshine durations are observed in Samedan and Segl-Maria (54% to 57%), located in the high-lying Engadin Valley. This region is known for its exceptional sun exposure, as the main Alpine ridge to the south often blocks moist air masses and clouds originating from Italy.

2.1.3 Results

Based on the evaluated climatic properties, Davos provides the most representative conditions for the Silvretta Glacier, particularly regarding precipitation and the relative duration of sunshine. For this reason, the Davos station is selected to obtain reference climate data for the glacier.

Davos is also geographically the closest station to the Silvretta Glacier compared to the other locations considered. The elevation difference between Davos and the stations in the Engadin Valley (Samedan and Segl-Maria) is relatively small, while the difference to the glacier itself is considerably larger.

Although no single station can perfectly capture the climate across the full altitude range of the glacier, the proximity, elevation, and observed climate data from Davos make it a suitable choice for representing the glacier's conditions. This representation can be further refined by adjusting the measured temperatures to the mean altitude of the glacier. Therefore, the homogenized series of precipitation and temperature measurements from the Davos station will be used for the following analysis.

The homogenized monthly temperature and precipitation series for Davos were retrieved from

the MeteoSwiss data portal [8]. Homogenization corrects for non-climatic breaks (relocations, instrument changes), which is essential for century-scale trend analysis and for a consistent model calibration and reconstruction.

3 Melt model calibration

3.1 Question 3.1:

What was your procedure to calibrate the DDF? Make sure to provide a step by step explanation and to mention any assumption.

3.1.1 Fundamentals

In this task, the Degree Day Factor (DDF) is determined using a temperature index model. The main assumption of this model is that glacier melt (ablation) depends solely on air temperature, neglecting effects such as radiation, debris cover, or wind. Although this is a simplification of reality, it provides meaningful estimates when properly calibrated.

3.1.2 Methodology

The calibration procedure is based on the overall glacier mass balance between October 1994 and October 2022. The following steps outline the approach in detail:

1. Selection of the model approach

Given the available data, two DEMs and meteorological observations from the Davos station, a lumped model was used rather than a distributed one. This means that the glacier is treated as a single unit with uniform conditions, instead of solving the mass balance for each grid cell. This simplification follows from the calculation of the mean annual glacier mass balance (calculated in Task 1), which represents an average value for the entire glacier. For a distributed model, however, the mass balance equation would have to be solved for each grid cell, substantially increasing the computational needs. Both DEMs represent the glacier surface in October, ensuring a consistent temporal reference. Furthermore, it is assumed that the glacier surface corresponds to the DEM of 2022 and remains constant throughout the considered period. The general mass balance equation is:

$$B = C - A \quad (3.1)$$

where B is the total mass balance, C the total accumulation and A the total ablation in the observed period.

2. Temperature correction for elevation difference

The meteorological data is obtained from the station at Davos (see Task 2) [8], which is located at a lower elevation than the glacier. To estimate the glacial temperature, the temperature data was corrected for the altitude difference using a temperature lapse rate, given in the task description [4], of

$$\frac{dT}{dz} = 6.5^\circ\text{C per km} \quad (3.2)$$

The mean glacier elevation was determined from the DEMs as $z_{\text{glacier,mean}} = 2773$ m a.s.l., while the weather station is located at $z_{\text{Davos}} = 1594$ m a.s.l.. The temperature correction is thus:

$$\Delta T = \frac{dT}{dz} \cdot (z_{\text{glacier,mean}} - z_{\text{Davos}}) \quad (3.3)$$

The corrected glacier temperature is therefore the following:

$$T_{\text{glacier}} = T_{\text{Davos}} - \Delta T \quad (3.4)$$

3. Calculation of the glacier mass balance

The total glacier mass balance for the period from 1994 to 2022 was obtained from the specific mass change rate calculated in Task 1. By multiplying the mean annual mass change by the number of years between the two DEMs, the total glacier mass change over the observation period is obtained in meters of water equivalent.

$$B_{1994-2022} = -0.9448 \text{ m w.e. / a} \cdot \text{years} \quad (3.5)$$

4. Estimation of accumulation (C)

The accumulation represents all processes adding mass to the glacier, mainly snowfall. It was calculated using a simplified version, the "block rain model":

$$C = P_{\text{ref}} \cdot r_s \quad (3.6)$$

In the original formulation [5], the model also considers corrections for gauge undercatch, a precipitation lapse rate, and snow redistribution effects:

$$C = P_{\text{ref}} \cdot r_s \cdot (1 + c_{\text{prec}}) \cdot [1 + (z - z_{\text{ref}}) \cdot \frac{dP}{dz}] \cdot D_{\text{snow}} \quad (3.7)$$

However, since detailed data for these parameters were not available, several simplifying assumptions were applied. The correction for gauge undercatch was neglected ($c_{\text{prec}} = 0$), the vertical gradient of precipitation was set to zero ($\frac{dP}{dz} = 0$), and snow redistribution processes were ignored ($D_{\text{snow}} = 1$). With these assumptions, the model simplifies to Equation 3.6 [5]. Here, P_{ref} denotes the total monthly precipitation measured at the reference station in Davos (see Task 2), and r_s is the fraction of solid precipitation. To distinguish between snow and rain, a temperature threshold of $T_0 = 2^\circ\text{C}$ was applied. For months where the mean glacier temperature was below this threshold, it was assumed that all precipitation would fall as snow ($r_s = 1$), otherwise $r_s = 0$. The monthly precipitation data was then summed up over the whole observation period to obtain the total accumulation $C_{1994-2022}$.

5. Calculation of ablation (A)

From the overall mass balance equation 3.1:

$$A_{1994-2022} = C_{1994-2022} - B_{1994-2022} \quad (3.8)$$

This value represents the total mass lost through melting in the considered period.

6. Computation of the total melting temperature (T)

The total positive temperature sum $T_{1994-2022}$ was determined by summing all monthly mean air temperatures, from start of October 1994, till end of September 2022, above the melting threshold $T_{\text{melt}} = 0^\circ\text{C}$. Each monthly mean temperature exceeding T_{melt} was multiplied by the average number of days per month (30.42 days) (see Task Description [4]), and the results were summed up over the entire period as following:

$$T_{1994-2022} = \sum_{T_i > 0^\circ\text{C}} T_i \cdot 30.42 \quad (3.9)$$

7. Calibration of the Degree Day Factor (DDF)

Finally, the DDF can be determined from:

$$A = DDF \cdot T \quad (3.10)$$

$$\text{for: } T > T_{\text{melt}} \quad (3.11)$$

which can be rearranged to:

$$DDF = \frac{A_{1994-2022}}{T_{1994-2022}} = \frac{C_{1994-2022} - B_{1994-2022}}{T_{1994-2022}} \quad (3.12)$$

The resulting DDF, which in the case of the employed lumped model is assumed to be independent of location and time over the glacier, expresses the amount of melt in millimeters of water equivalent per degree per day [5].

3.2 Question 3.2:

What value (and unit) do you obtain for the calibrated DDF, and how does it compare to values reported in the literature? Make sure to have "days" somewhere in the units for the "Degree Day Factor", and compare your result against at least two published DDF values (provide the corresponding references). Discuss what might be at the origin of the differences that you find (if any).

3.2.1 Results

The calibration procedure described in section 3.1 resulted in a Degree Day Factor of

$$DDF = 3.32 \text{ mm w.e. } ^\circ\text{C}^{-1} \text{ d}^{-1} \quad (3.13)$$

This value indicates that, for each day where the mean air temperature exceeds the melting threshold $T_{\text{melt}} = 0^\circ\text{C}$ by one degree, approximately 3.32 mm of water equivalent melt occurs on the glacier surface.

3.2.2 Discussion

In the paper from Braithwaite and Zhang [3], the sensitivity of glacier mass balance to temperature changes was analysed for five Swiss glaciers. The Silvretta Glacier, which is also the focus of this assignment, was included in this comparison. The authors compared the annual changes in mass balance resulting from an increase in temperature of 1°C , calculated both with a regression approach and by tuning a degree-day factor model. Their paper also contains a list of the various Degree Day Factors from previously published studies (see Figure 3.1). Comparing the resulting DDF from task 3.1 with the values from literature $DDF_{\text{ice}} = 5 - 7 \text{ mm w.e. } ^\circ\text{C}^{-1} \text{ d}^{-1}$ for various Swiss glaciers and $DDF_{\text{snow}} = 4.5 \text{ mm w.e. } ^\circ\text{C}^{-1} \text{ d}^{-1}$ for the Wiessfluhjoch, which is located directly above Davos, we can see that the resulting value is lower than both of these values.

This can be explained by several factors. First, the applied model is a highly simplified degree-day approach, which assumes that melt is linearly related to air temperature and neglects other components of the surface energy balance such as shortwave and longwave radiation, turbulent fluxes, and latent heat exchange. As a result, the model may underestimate total melt, especially during periods with high incoming radiation but moderate air temperatures.

Second, the DDF strongly depends on the surface type. Ice surfaces generally have higher DDF values than snow surfaces due to their lower albedo and higher absorption of solar radiation. Since the glacier surface in our study area was likely partially snow-covered during large parts of the observation period, the effective DDF is expected to be lower than the values reported in the literature for predominantly ice-covered glaciers.

Third, the long averaging period from 1994 to 2022 may have smoothed short-term melt extremes, leading to a smaller average ablation per degree and per day.

Finally, local climatic conditions also play an important role. The meteorological station in Davos, which was used as the reference for temperature and precipitation data, is located in a relatively cool and high-altitude valley with frequent shading and moderate solar radiation compared to more exposed glacier surfaces. These local effects reduce the available energy for melt and consequently lead to lower DDF values. Therefore, the lower DDF obtained here is physically reasonable given the simplified model formulation, the partial snow coverage, the long averaging period, and the cooler climatic conditions at Davos.

Ice	Snow	Location	Source
5.0–7.0	4.5	Weissfluhjoch, Switzerland	Zingg (1951)
18.6		Various Swiss glaciers	Kasser (1959)
13.8		EGIG ⁺ Camp IV, Greenland	Ambach (1963, 1988)
6.3		Spitsbergen	Schytt (1964)
		St. Supphellebreen, Norway	Orheim (1970)
5.5 ± 2.3	5.5	Gr. Aletschgletscher, Switzerland	Lang and others (1977)
		Various glaciers, Norway	Braithwaite (1977)
6.3 ± 1.0	4.5	Weissfluhjoch, Switzerland	De Quervain (1979)
20.1–22.2		Arctic Canada	Braithwaite (1981)
6.0		GIMEX [†] profile, West Greenland	Van de Wal (1992)
	3.0	Franz Josef Glacier, New Zealand	Woo and Fitzharris (1992)
7.7	5.7	Sátujökull, Iceland	Jóhannesson and others (1993)
6.4	4.4	Nigardsbreen, Norway	Jóhannesson and others (1993)
7.3	2.8	Qamanárssúp sermia, West Greenland	Jóhannesson and others (1993)
6.0	4.5	Alfotbreen, Norway	Laumann and Reeh (1993)
5.5	4.0	Nigardsbreen, Norway	Laumann and Reeh (1993)
5.5	3.5	Hellstugubreen, Norway	Laumann and Reeh (1993)
8.1	2.9	Nordbogletscher, West Greenland	Braithwaite (1995b)
8.3	3.7	Qamanárssúp sermia, West Greenland	Braithwaite (1995b)
6.9–7.1		Patagonia, Argentina	Takeuchi and others (1996)
5.9–9.8		North Greenland	Braithwaite and others (1998)
6.2	3.8	Glacier de Sarnes, France	Vincent and Vallon (1997)
8.3–9.4		Griesgletscher, Switzerland	This study
6.3	4.4	Storglaciren, Sweden	Hock (1999)

Figure 3.1: Positive degree-day factors at various location in mm w.e. °C⁻¹ d⁻¹ [3]

Of course, this raises the question of whether such a low DDF factor of 3.32 mm w.e. °C⁻¹ d⁻¹ is realistic at all. If we compare our value with the observed DDF values in western China in Figure 3.2, it becomes clear that several glaciers have a DDF value of less than 4 mm w.e. °C⁻¹ d⁻¹. However, most of these are located at an altitude of above 4,000 metres. The Batura Glacier (2780 m a.s.l.) on the other hand, is at an altitude similar to that of the Silvretta Glacier (2773 m a.s.l.) and reaches a value of 3.4 mm w.e. °C⁻¹ d⁻¹. Furthermore, the glaciers analysed by Zhang et al.[10] were observed over relatively short periods of only a few weeks to months, typically during the peak melt season. In contrast, our calibration in Task 3.1 is based on a 28 year period. Short term studies tend to capture more extreme melt conditions, resulting in higher

DDF values, whereas long term analyses include colder years and periods with lower radiation, which reduce the average ablation per degree day. Additionally, recent work by Braithwaite and Hughes [2] suggests that DDFs have increased with ongoing climate change, meaning that shorter, more recent time series generally yield higher values than those obtained over our 28 year observation period.

Mountain system	Glacier	DDF _{ice} mm d ⁻¹ °C ⁻¹	DDF _{snow} mm d ⁻¹ °C ⁻¹	Altitude m a.s.l.	Period	Source
Tien Shan	Ürümqi No.1	8.5		3831–3945	1986–93	Liu and others (1996)
		7.3		3754–3898	1986–88	Liu and others (1996)
			3.1	4048	1986–93	Liu and others (1996)
	Keqicar Baqi	4.5		3347	28 Jun–12 Sep 2003	Zhang and others (2005)
		7.0		4216	11 Jul–13 Sep 2003	Zhang and others (2005)
	Qiongtailan	4.5		3675	17 Jun–31 Jul 1978	This study
		7.3		4100	25 Jun–14 Aug 1978	This study
		8.6		4200	21 Jun–31 Jul 1978	This study
			3.4	4400	21 Jun–11 Aug 1978	This study
				3301	24 Aug 1982–Aug 1983	This study
Hengduan mountains	Baishuihe No.1	5.0		4600	23 Jun–30 Aug 1982	This study
		13.3		4800	26 Jun–11 Jul 1982	This study
			5.9	4540	20 Sep 1982–22 Sep 1983	This study
	Dagongba	13.2		4550	15 Jul 1982–15 Jul 1983	This study
	Xiaogongba	12.0				
Karakoram mountains	Batura	3.4		2780	Jun–Aug 1975	This study
	Teram Kangri	5.9		4630	25 Jun–7 Sep 1987	This study
		6.4		4650	24 Jun–7 Sep 1987	This study
	Qirbulake	2.6		4750	6 Jun–30 Jul 1960	This study
	Yangbulake	4.3		4800	1–5 Jul 1987	This study
Kunlun Shan	Meikuang	3.0		4840	7 May–7 Sep 1989	This study
	Halong	4.7		4616	15–28 Jun 1981	This study
		3.6		4900	14–27 Jun 1981	This study
Tanggula mountains	Xiaodongkemadi	13.8		5425–5475	Jul–Aug 1993	Kayastha and others (2003)
Qilian Shan	Qiyi	7.2		4305–4619	Jul–Aug 2002	Kayastha and others (2003)
Himalaya	Kangwure	9.0		5700–6000	20 Jul–25 Aug 1993	This study

Figure 3.2: DDFs on different monitored glaciers in Western China [10]

3.3 Question 3.3:

Reflect upon the various assumptions that were necessary for the calibration. Which assumptions are likely to introduce a bias? In what direction do the individual biases go, and which assumptions are likely to introduce the largest biases?

During the calibration of the Degree Day Factor for the Silvretta Glacier, several simplifications and assumptions were necessary, each of which may introduce biases into the results.

1. Simplified DDF model

The model assumes that ablation is directly proportional to air temperature. While temperature is indeed a main driver of melting, this simplification ignores other important influences, such as local topography, for example shading, differences in surface albedo between snow and ice, and variations in solar radiation or cloud cover. These omissions could either over- or underestimate actual melt depending on the specific conditions at different parts of the glacier.

2. Temperature data and lapse rate

Air temperatures were derived from the closest meteorological station in Davos and ad-

justed using a lapse rate of $\frac{dT}{dz} = 6.5 \text{ }^\circ\text{C km}^{-1}$. In reality, temperature variations across the glacier are influenced by local factors such as wind, shading, or elevation-dependent inversion layers. Using a uniform lapse rate may overestimate temperatures at higher elevations, leading to underestimated snow accumulation, while underestimating temperatures at lower elevations could result in overestimated ablation.

3. Precipitation and accumulation

Accumulation was estimated from snowfall data, neglecting contributions from avalanches, wind-blown snow, or hoar deposition. Additionally, the block rain model was used to sum monthly precipitation and distinguish snow from rain by a fixed temperature threshold ($T_0 = 2^\circ\text{C}$). This approach may overestimate accumulation when rain is misclassified as snow or when snowfall occurs in short, intense events that are not captured by the monthly average. Overall, these factors likely lead to a slight underestimation of total accumulation.

4. Threshold temperature for melting

The model assumes that melting occurs only when air temperature exceeds 0°C . In reality, melting can occur at slightly lower temperatures under sunny or wind-exposed conditions, and sometimes melt does not occur even above 0°C under cloudy or sheltered conditions. This simplification introduces some uncertainty, but its net bias is difficult to quantify and may partially cancel out over the year.

5. Glacier geometry and area

The glacier surface was represented by the DEM from 2022. This neglects temporal changes in glacier elevation and area due to ice loss. As the glacier thins and retreats over time the total ice volume decreases, but the glacier surface per volume can increase because the glacier surface declines more slowly than its volume. This evolution of the volume-to-surface ratio is not accounted for in the model, likely leading to an overestimation of melt in the years before 2022 and representing the largest source of bias in the DDF calibration.

In general, the combination of assumptions can either increase or decrease the estimated DDF. Some biases, such as neglecting wind-blown snow, tend to reduce accumulation and increase the DDF, whereas assuming a fixed glacier surface may underestimate melt and decrease the DDF. Among all assumptions, using a constant DEM for a dynamic glacier surface is likely the most significant source of bias. Despite these simplifications, the calibrated DDF provides a reasonable estimate, but a more detailed, distributed model would be required to refine it further for the Silvretta Glacier.

4 Mass balance reconstruction

4.1 Question 4:

What are the (i) cumulative and (ii) average glacier mass balance (in m w.e. and m w.e. a^{-1} , respectively) and what do you observe in the yearly time series? Check whether you can discern trends, and whether there are periods or individual years that stand out.

4.1.1 Fundamentals

The meteorological data from 1935 to 2024 can now be used as an input for the calibrated model from Task 3.1 to reconstruct the mass balance of the Silvretta Glacier. Accumulation (C) and ablation (A) of the glacier were calculated according to the equations 3.6 and 3.11. With the calculated values of the accumulation and ablation the glacier mass balance (B) can be computed according to equation 3.1. The following assumptions were made to simplify the calculation of the glacier mass balance:

1. The calculations are based on the hydrological year (01.October – 30.September).
2. One month is assumed to have an average length of 30.42 days.
3. The DDF values are taken from Section 3.1.
4. The monthly mean temperature at the glacier T_{glacier} is calculated according to Equation 3.4.
5. When the air temperature is below 2°C , precipitation leads to accumulation.
6. When the air temperature is above 0°C , ablation occurs.
7. No change in glacier geometry is considered. Therefore, the glacier area from the year 2022 is used.

4.1.2 Results

Figure 4.1 depicts the evolution of the glacier mass balance from 1935 to 2024, including both the mean annual mass balance over the entire observation period and a quadratic trend line. The cumulative glacier mass balance is shown in Figure 4.2. It illustrates how the total variation in glacier mass has changed over the years. The values show the total change in the mass of the glacier up to the respective year in relation to the reference year 1935, when the measurement series began. The total cumulative (from 1935 to 2024) and mean glacier mass balance results are summarized in Table 4.1.

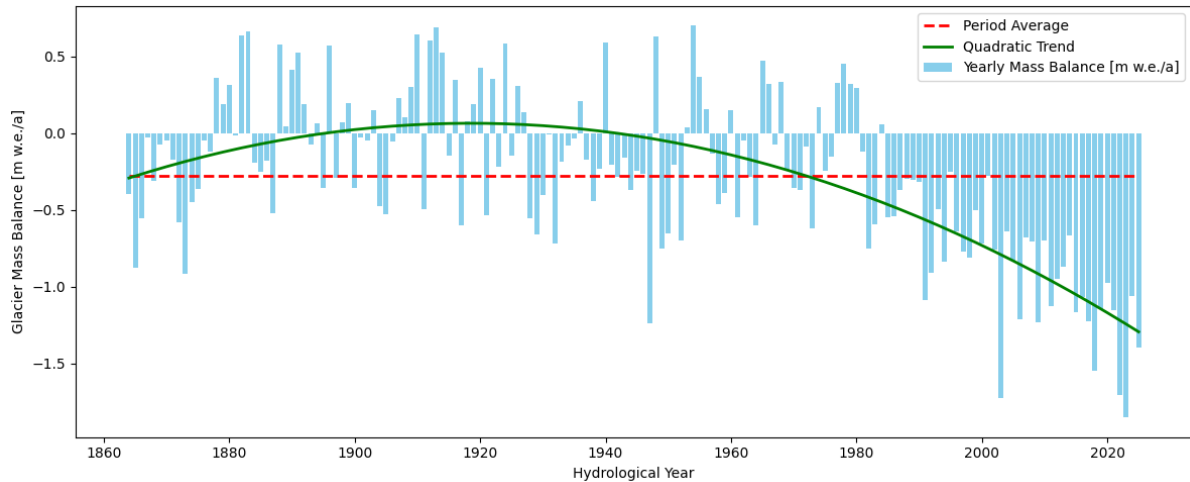


Figure 4.1: Glacier mass balance (1935-2024)

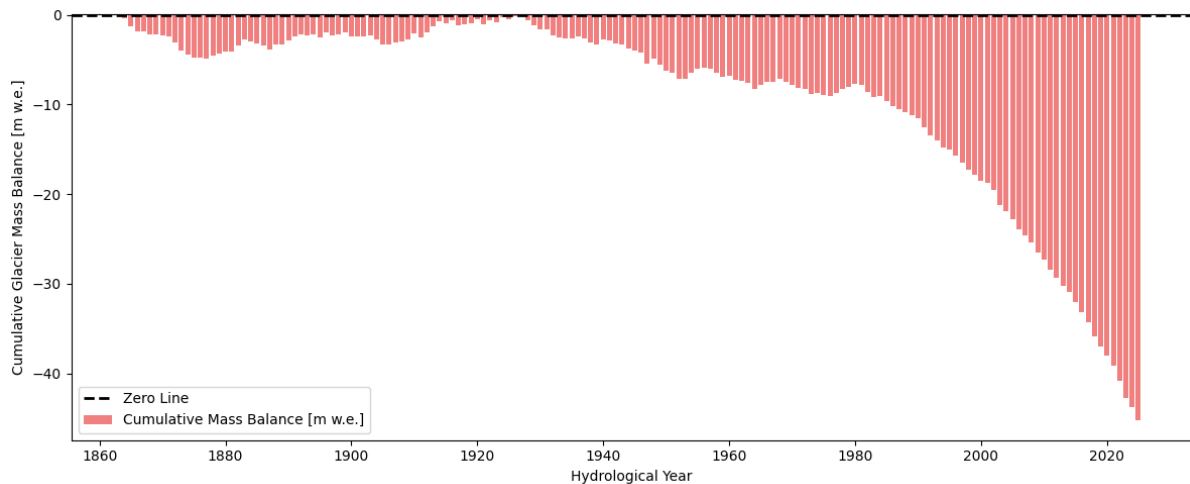


Figure 4.2: Cumulative glacier mass balance (1935-2024)

Cumulative mass balance (1935–2024)	-45.16	m w.e.
Average annual mass balance (1935–2024)	-0.28	m w.e. a ⁻¹

Table 4.1: Results of the reconstructed glacier mass balance

4.1.3 Interpretation

The reconstructed glacier mass balance exhibits a clear and persistent long-term decline. Over the full observation period (1935–2024), the mean annual mass balance is approx. $-0.28 \text{ m w.e. a}^{-1}$, and by 2024 the cumulative mass balance amounts to -45.16 m w.e. (Table 4.1). This sustained negative balance indicates that ablation has systematically exceeded accumulation, implying a progressive reduction of glacier volume and surface area.

Figure 4.1 helps to interpret the temporal evolution of this signal. The red horizontal line denotes the period average at $-0.28 \text{ m w.e. a}^{-1}$. In several recent decades, lower values were

measured, indicating periods of significantly accelerated glacier melt. The green quadratic trend line captures the non-linear character of the record. A near-neutral to slightly positive tendency in the earlier part of the series is followed by a marked downturn beginning in the late 20th century and accelerating thereafter. The curvature of the trend therefore indicates an intensification of glacier mass loss in recent decades, consistent with a strengthened climatic forcing.

The cumulative perspective in Figure 4.2 reinforces this interpretation. After remaining close to zero for part of the mid-20th century, the cumulative curve turns persistently negative and steepens from around 1980 onward, with no multi-year recovery phases of comparable magnitude. This behavior reflects that occasional years with near-zero or slightly positive balances were insufficient to offset the sequence of strongly negative years in the last 50 years.

Individual extremes further illustrate the recent acceleration. Exceptionally warm hydrological years such as 2003 and 2023 recorded large negative balances on the order of -1.72 and -1.85 m w.e., respectively, whereas the largest loss prior to 1980 occurred in 1947 (about -1.24 m w.e.). These events, together with the downward curvature of the trend line and the steadily declining cumulative curve, point to a regime of enhanced ablation dominating the contemporary period.

Given the magnitude, persistence, and acceleration of losses evident in Figures 4.1 and 4.2, a spontaneous return to neutral or positive mass balance appears unlikely without substantial and sustained changes in environmental conditions. While the reconstruction is subject to methodological assumptions (e.g., fixed glacier area, temperature thresholds for accumulation and ablation) and thus carries uncertainty in absolute magnitudes, the convergence of the period average, the quadratic trend, and the cumulative trajectory provides robust evidence for a significant and continuing reduction in glacier mass over 1935–2024.

The reconstructed 1935–2024 series shows a persistent negative trend with an acceleration of losses beginning in the 1980s, consistent with regional alpine warming. The pattern aligns with the observed, accelerated retreat of alpine glaciers and indicates increasing frequency of years with strongly negative mass balance.

5 Future projection

5.1 Question 5.1:

What was the glacier volume in 2022? Express your results in km³.

The glacier volume in 2022 was estimated using the volume-area scaling method [6], following this formula:

$$V_{2022} = c \cdot A_{2022}^{\gamma} = 0.1041 \text{ km}^3 \quad (5.1)$$

where $c = 0.033$ and $\gamma = 1.36$ for glaciers [1] and the glacial area in 2022 was around $A_{2022} = 2.327 \text{ km}^2$ (see Section 1.1.3).

5.2 Question 5.2:

Based on your projection, when (year) will the glacier have lost 50% of the volume it had in 2022? Discuss whether the resulting "projection" is likely to be an under- or overestimate, and why.

5.2.1 Results

First, half the glacier volume is determined with the following calculation:

$$V_{2022, 50\%} = 0.5 \cdot V_{2022} = 0.5 \cdot 0.1041 \text{ km}^3 = 0.0520 \text{ km}^3 \quad (5.2)$$

It is assumed that the glacier mass balance rate will remain constant like the average over the years from 1994 to 2024 and the area will remain constant over the period considered at its value in 2022 [4].

$$B_{\text{avg}, 1994-2024} = -0.97 \text{ m w.e. / a} \quad (5.3)$$

The number of years n required for a reduction of the glacier to half its volume in the year 2022 can be calculated as follows:

$$n = \frac{V_{2022, 50\%}}{B_{\text{avg}, 1994-2024} \cdot A_{2022}} \cdot \frac{\rho_{\text{glacier}}}{\rho_{\text{water}}} = 21.17 \text{ a} \quad (5.4)$$

where $\rho_{\text{glacier}} = 900 \text{ kg/km}^3$ and $\rho_{\text{water}} = 1000 \text{ kg/km}^3$.

5.2.2 Discussion

The projection of future glacier volume presented in this study is based on simplified assumptions that introduce uncertainties into the results. The glacier volume in 2022 was estimated as $V_{2022} =$

0.1041 km^3 . Assuming a constant mean mass balance rate of $B_{\text{avg}, 1994-2024} = -0.97 \text{ m w.e./a}$ and a constant glacier area of $A_{2022} = 2.327 \text{ km}^2$, the glacier is projected to lose 50% of its 2022 volume after approximately 21.17 years, corresponding to the year 2044.

This projection is subject to several limitations. The assumption of a constant glacier area is a major simplification, whose implications are outlined in 3.3. Additionally, as the glacier retreats into higher elevations, total annual mass loss will decrease due to lower temperatures and thereby slowing the rate of volume reduction. Consequently, the actual time required to reach 50% of the 2022 volume is likely to be longer than calculated, meaning the threshold will occur later than 2044.

The assumption of a constant mass balance rate further limits the accuracy of the projection. The applied mean value is derived from observations between 1994 and 2024 and assumes stable climatic conditions. Future changes in temperature and precipitation may significantly affect the mass balance, leading to deviations from the estimated rate and thus altering the timing of volume loss.

Additionally, the use of the volume–area scaling relationship introduces uncertainty. Although widely used in glaciology, this empirical approach is an approximation, and deviations from the applied scaling exponent $\gamma = 1.36$ or coefficient $c = 0.033$ could lead to differences between the estimated and actual glacier volume.

Overall, the result that the glacier will lose 50% of its 2022 volume by 2044 should be considered a first-order approximation. When a glacier shrinks, there are opposing mechanisms influencing how fast it loses mass. A smaller surface area tends to slow melting because less ice is exposed to atmospheric heat. However, a smaller volume-to-area ratio increases relative exposure, which accelerates melting. At the same time, as the glacier retreats to higher elevations, cooler temperatures and increased snowfall (per glacier volume) improve its mass balance and slow down melting. Therefore, the net effect on the response time depends on which process dominates. A more accurate projection would require a time-dependent treatment of glacier area, variable mass balance scenarios based on climate projections, and refined scaling parameters.

5.3 Question 5.3:

Can you think of a strategy that avoids the simplification that the glacier area remains constant at the 2022 level? Explain how you could avoid this assumption of a constant area, and what additional information, tools, or methods (if any) you would need for that.

A more accurate prediction of future glacier volume is possible if time-dependent observations are used instead of assuming a constant glacier area. Satellite images from missions like Sentinel or Landsat provide detailed yearly data on glacier extent. These images can be analyzed in GIS software to determine the area for each year. By including this change in area over time in the volume–area relationship, systematic errors are reduced, and the estimation of volume change becomes more realistic.

To further improve accuracy, the ice thickness should also be considered. Measurements with Ground Penetrating Radar (GPR) can directly determine ice thickness and thus complement the volume estimates derived from area data. When these measurements are combined with time series of mass balance data, a dynamic model of glacier evolution can be created that reflects

both geometric changes and climatic influences.

Finally, all these data can be integrated into a numerical model to simulate future glacier changes under different climate scenarios. This model takes into account feedbacks between shrinking area, changing mass balance, and volume loss, avoiding the oversimplification caused by assuming a constant glacier area.

Bibliography

- [1] BAHR, D. B.: Estimation of Glacier Volume and Volume Change by Scaling Methods. Version: 2011. http://dx.doi.org/10.1007/978-90-481-2642-2_690. In: SINGH, V. P. (Hrsg.) ; SINGH, P. (Hrsg.) ; HARITASHYA, U. K. (Hrsg.): *Encyclopedia of Snow, Ice and Glaciers*. Dordrecht : Springer, 2011 (Encyclopedia of Earth Sciences Series). – DOI 10.1007/978-90-481-2642-2_690
- [2] BRAITHWAITE, R. J. ; HUGHES, P. D.: *Positive degree-day sums in the Alps: a direct link between glacier melt and international climate policy*. <https://www.cambridge.org/core/journals/journal-of-glaciology/article/positive-degreeday-sums-in-the-alps-a-direct-link-between-glacier-melt-and-international-climate-policy/BB3749443F47294C31155F46C2D67B00>, 2022
- [3] BRAITHWAITE, R. J. ; ZHANG, Y.: *Sensitivity of mass balance of five Swiss glaciers to temperature changes assessed by tuning a degree-day model*. <https://www.cambridge.org/core/journals/journal-of-glaciology/article/sensitivity-of-mass-balance-of-five-swiss-glaciers-to-temperature-changes-assessed-by-tuning-a-degreeday-model/A8FEFF6E7227194CE7F977A51FA30B9A>, 2000
- [4] FARINOTTI, Prof. Dr. D.: *Applied Glaciology – Assignments part 1 Fall Semester 2025*. 2025. – Assignment for Applied Glaciology Lecture
- [5] FARINOTTI, Prof. Dr. D.: *appliedglaciology02glaciologyandhydropower2025part1*. 2025. – Slides
- [6] FARINOTTI, Prof. Dr. D.: *appliedglaciology02glaciologyandhydropower2025part2*. 2025. – Slides
- [7] METEOSCHWEIZ: *Messwerte und Messnetze*. <https://www.meteoschweiz.admin.ch/service-und-publikationen/applikationen/messwerte-und-messnetze.html#param=messnetz-klima&table=false>, . – Accessed: 2025-10-08
- [8] METEOSWISS: *Homogenized Climate Data Tables: Davos (NBCN Station)*. <https://www.meteoswiss.admin.ch/services-and-publications/applications/ext/climate-tables-homogenized.html#https%3A%2F%2Fservice.meteoswiss.ch%2Fproductbrowser%2FproductDisplay%2Fclimate-tables-homogenized%3Flang=en&cg1.location=DAV&cg1.productName=climate-reports-tables-homogenized>, n.d.. – Accessed: 2025-10-08
- [9] MITTELHOLZER, WALTER: *Silvrettagletscher, Fluchthorn, Piz Buin v. W. aus 2800 m*. <https://ba.e-pics.ethz.ch/#detail-asset=fcbe79e8-f3f5-4dbf-acfa-6cfc07794415>, 1925. – Accessed: 2025-10-19
- [10] ZHANG, Y. ; LIU, S. ; DING, Y.: *Observed degree-day factors and their spatial variation on glaciers in western China*. <https://www.cambridge.org/core/journals/annals-of-glaciology/article/observed-degreeday-factors-and-their-spatial-variation-on-glaciers-in-western-china/688A42CBB828EAD3EOA55D55748F1C56>, 2006

A Appendix

A.1 Task 1

```

1     import numpy as np
2     import os
3     os.chdir(os.path.dirname(__file__))
4
5     def read_asc_file(file_path):
6         with open(file_path, 'r') as file:
7             header = []
8             #Save Values from Header
9             for _ in range(6):
10                 header.append(file.readline().strip())
11
12             cellsize = float(header[4].split()[1])
13             nodata_value = float(header[5].split()[1])
14
15             data = np.loadtxt(file, dtype=float)
16             data[data == nodata_value] = np.nan
17
18         return data, cellsize
19
20     def calculate_volume_and_mass_change(dem_1994, dem_2022, mask_1994,
21                                         mask_2022, cell_area, output_file=None):
22         height_diff = dem_2022 - dem_1994
23
24         glacier_mask_1994 = (mask_1994 == 1)
25         glacier_mask_2022 = (mask_2022 == 1)
26         combined_glacier_mask = glacier_mask_1994 | glacier_mask_2022
27
28         print('Average_elevation: ' + str(round((np.mean(dem_1994[
29             glacier_mask_1994]) + np.mean(dem_2022[glacier_mask_2022]))
30             * 0.5, 2)) + 'm') #average elevation (for lapse rate
31                               conversion)
32         print('Average_elevation_1994: ' + str(round(np.mean(dem_1994[
33             glacier_mask_1994]), 2)) + 'm') #average elevation (for lapse
34             rate conversion)
35         print('Average_elevation_2022: ' + str(round(np.mean(dem_2022[
36             glacier_mask_2022]), 2)) + 'm') #average elevation (for lapse
37             rate conversion)
38         print('Min_elevation_1994: ' + str(np.min(dem_1994[
39             glacier_mask_1994])) + 'm')
40         print('Max_elevation_1994: ' + str(np.max(dem_1994[
41             glacier_mask_1994])) + 'm')
42         print('Min_elevation_2022: ' + str(np.min(dem_2022[
43             glacier_mask_2022])) + 'm')
44         print('Max_elevation_2022: ' + str(np.max(dem_2022[
45             glacier_mask_2022])) + 'm')
46
47         glacier_height_diff = height_diff[combined_glacier_mask]
48         glacier_height_diff_array = np.where(combined_glacier_mask,
49             height_diff, 0)

```

```

38
39     if output_file:
40         np.savetxt(output_file, glacier_height_diff_array,
41                     delimiter=",", fmt="%.2f")
42
43     volume_change = np.sum(glacier_height_diff) * cell_area
44
45     ice_density = 850
46     mass_change = volume_change * ice_density
47
48     glacier_area_1994 = np.sum(glacier_mask_1994) * cell_area
49     glacier_area_2022 = np.sum(glacier_mask_2022) * cell_area
50     print(f"A_1994: {glacier_area_1994:.2f} m2")
51     print(f"A_2022: {glacier_area_2022:.2f} m2")
52
53     average_glacier_area = (glacier_area_1994 + glacier_area_2022) /
54                             2
55     print(f"A_Avg: {average_glacier_area:.2f} m2")
56
57     years = 2022 - 1994
58     print("Time Period: " + str(years))
59     water_density = 1000
60     specific_mass_change_rate = (mass_change / (water_density *
61         average_glacier_area)) / years
62
63     return volume_change, mass_change, average_glacier_area,
64         specific_mass_change_rate
65
66 def main():
67     dem_1994, cellsize = read_asc_file('dem_silvretta_1994.asc')
68     dem_2022, _ = read_asc_file('dem_silvretta_2022.asc')
69     mask_1994, _ = read_asc_file('glacier_mask_silvretta_1994.asc')
70     mask_2022, _ = read_asc_file('glacier_mask_silvretta_2022.asc')
71     #print(cellsize)
72     #returns amount of cells of data
73     print("Shape of Grid for Glacier: " + str(dem_1994.shape[0]) + "
74         Rows and " + str(dem_1994.shape[1]) + " Columns")
75
76     cell_area = cellsize ** 2
77     print("Cell Area: " + str(cell_area))
78
79     output_csv = "glacier_height_diff_array.csv"
80
81     volume_change, mass_change, average_glacier_area,
82     specific_mass_change_rate = calculate_volume_and_mass_change(
83         dem_1994, dem_2022, mask_1994, mask_2022, cell_area,
84         output_file=output_csv)
85
86     print(f"volume change: {volume_change:.2f} m3")
87     print(f"mass change: {mass_change:.2f} kg")
88     print(f"average area: {average_glacier_area:.2f} m2")
89     print(f"specific mass change rate: {specific_mass_change_rate:.4
90         f} mwe/a")
91
92 if __name__ == "__main__":
93     main()

```

A.2 Task 3

```

1 import os
2 import numpy as np
3 import pandas as pd
4 os.chdir(os.path.dirname(__file__))
5
6
7 def read_meteoswiss_txt(file_path):
8     with open(file_path, 'r') as file:
9
10         # --- Read header lines until reaching the table header
11         ---
12         for line in file:
13             if line.startswith('Altitude'):
14                 Alt = int(''.join(filter(str.isdigit,
15                                         line)))
16             if line.startswith("Year"): # Table header
17                 starts with "Year"
18                 break
19
20         # --- Read the remaining lines (the data table) ---
21         data = pd.read_csv(
22             file,
23             delim_whitespace=True,
24             na_values=["NA"],
25             names=["Year", "Month", "Temperature", "
26                   Precipitation"],
27         )
28
29     return data, Alt
30
31 def main():
32     Data, Station_Altitude = read_meteoswiss_txt('climate-reports-
33     tables-homogenized_DAV.txt')
34     print(Data)
35     Data_1994_2022 = Data[((Data['Year'] == 1994) & (Data['Month']
36     >= 10)) | ((Data['Year'] == 2022) & (Data['Month'] < 10)) |
37     ((Data['Year'] > 1994) & (Data['Year'] < 2022))].copy()
38
39     temperature_lapse_rate = 6.5 #°Ckm-1
40     B = -0.9448 #mwe/a
41     years = 28
42     B_tot = B*years
43     Avg_Glacier_Elevation = 2777.43 #m
44
45     Data_1994_2022['T_Glacier'] = Data_1994_2022['Temperature'] -
46     temperature_lapse_rate*(Avg_Glacier_Elevation -
47     Station_Altitude)/1000
48     Data_1994_2022['r_s'] = (Data_1994_2022['T_Glacier'] < 2).astype
49     (int)
50     Data_1994_2022['C[mwe]'] = Data_1994_2022['Precipitation']*
51     Data_1994_2022['r_s']*0.001
52     Data_1994_2022['Melt_Mask'] = (Data_1994_2022['T_Glacier'] > 0).
53     astype(int)

```

```

43 Data_1994_2022['T[°C*d]'] = abs(Data_1994_2022['T_Glacier']*
44     Data_1994_2022['Melt_Mask']*30.42)
45
46 C_tot = Data_1994_2022['C[mwe]'].sum() #m w.e.
47 A_tot = C_tot - B_tot #m w.e.
48 T_tot = Data_1994_2022['T[°C*d]'].sum() #°C*d
49 DDF = A_tot/T_tot*1000 #mm w.e./(°C*d)
50
51 print('C_tot:␣' + str(round(C_tot, 2)))
52 print('A_tot:␣' + str(round(A_tot, 2)) + '␣mmw.e.')
53 print('T_tot:␣' + str(round(T_tot,2)) + '␣°C*d')
54 print('DDF:␣' + str(round(DDF,2)) + '␣mmw.e./(°C*d)')
55
56
57
58
59
60
61 if __name__ == "__main__":
62     main()

```

A.3 Task 4 & 5

```

1 # --- Plot 1: Yearly Mass Balance ---
2 x = annual['HydroYear'].values
3 y = annual['B[mwe/a]'].values
4
5 # Period average
6 avg = np.mean(y)
7
8 # Quadratic trendline fit
9 coeffs = np.polyfit(x, y, 2) # 2. Ordnung
10 trend = np.polyval(coeffs, x)
11
12 plt.figure(figsize=(12,5))
13 plt.bar(x, y, color='skyblue', label='Yearly␣Mass␣Balance␣[m␣w.e
14     ./a]')
15 plt.plot(x, [avg]*len(x), color='red', linestyle='--', linewidth
16     =2, label='Period␣Average')
17 plt.plot(x, trend, color='green', linestyle='--', linewidth=2,
18     label='Quadratic␣Trend')
19 plt.xlabel('Hydrological␣Year')
20 plt.ylabel('Glacier␣Mass␣Balance␣[m␣w.e./a]')
21 plt.legend()
22 plt.tight_layout()
23 plt.show()
24
25 # --- Plot 2: Cumulative Mass Balance ---
26 cum_y = annual['Cumulative_B[mwe]'].values
27
28 plt.figure(figsize=(12,5))
29 plt.bar(x, cum_y, color='lightcoral', label='Cumulative␣Mass␣
30     Balance␣[m␣w.e.]')

```

```
27 plt.axhline(0, color='black', linestyle='--', linewidth=2, label
28           ='Zero_Line')
29 plt.xlabel('Hydrological_Year')
30 plt.ylabel('Cumulative_Glacier_Mass_Balance_[m_w.e.]')
31 plt.legend()
32 plt.tight_layout()
33 plt.show()
34 # --- Output of data needed for Tasks 4 and 5 ---
35 print(f"Cumulative_mass_balance_1935_2024 :_{annual['B[mwe/a]'].sum():.2f}_m_w.e.")
36 print(f"Average_annual_mass_balance:_{annual['B[mwe/a]'].mean():.2f}_m_w.e./a")
37 print(f"Average_annual_mass_balance_(1994-2024) :_{annual.loc[(
38     annual['HydroYear']>=1994), 'B[mwe/a]'].mean():.2f}_m_w.e./a")
39
40 for year in [1947, 2003, 2023]:
41     if year in annual['HydroYear'].values:
42         value = annual.loc[annual['HydroYear'] == year,
43                             'B[mwe/a]'].values[0]
44         print(f"Year_{year}:_Glacier_mass_balance=_{
45             value:.2f}_m_w.e./a")
46     else:
47         print(f"Year_{year}:_No_data_available")
```